

TED(2010) 5066

(Revision -2010)

FIFTH SEMESTER DIPLOMA EXAMINATION IN INFORMATION TECHNOLOGY

COMPUTER GRAPHICS

(Model Question Paper)

(Maximum Marks: 100)

Time: 3 Hrs

Marks : 100

PART- A

I Answer the following questions in one or two sentences. Each question carries 2 marks.

1. List any two line drawing algorithms.
2. Define view port.
3. List main 3-D display methods.
4. State any four computer animation languages.
5. List the steps for animation sequence.

[5 x 2 = 10 marks]

PART –B

II Answer any five of the following questions. Each question carries 6 marks.

1. Illustrate Bresenham's circle drawing algorithm with an example.
2. Write the transformation matrix for the reflection of an object about the x axis. Line $y=0$.
3. List any three types of curve generation. Describe B-spline method for curve generation.
4. Describe 3-D translation along with its equation.
5. Suggest a method for specifying motion of objects in animation system. Explain.
6. Describe briefly about the raster-scan displays.
7. Explain briefly an algorithm for polygon clipping.

[5 x 6 =30 marks]

PART – C

(Answer one full question from each unit. Each question carries 15 marks.)

UNIT- I

III

1. Write DDA line drawing algorithm. [7]
2. Compare raster scan & random scan displays. [8]

OR

IV

1. Write C code for Bresenham's line drawing algorithm [7]
2. Explain the working principle of any two graphical I/O devices. [8]

UNIT- II

V

1. Describe the 2-D transformation matrix for rotation about arbitrary point. [6]
2. Explain briefly about any one of the line clipping algorithms. [6]
3. Write the basic geometric transformations. [3]

OR

VI

1. Explain 2-dimensional viewing transformation pipeline. [8]
2. With the help of a diagram, explain window-to-viewport co-ordinate Transformation. [7]

UNIT- III

VII

1. Describe 3-D scaling along with its matrix representation [8]
2. Summarize the properties of Bezier curves. [7]

OR

VIII

1. Describe B-spline curve with a suitable example [8]
2. Explain 3-D viewing with an example. [7]

UNIT- IV

IX

1. Suggest a suitable image-space method to detect visible surfaces. State its merits. [7]
2. Summarise the steps evolved in depth – buffer algorithm [5]
3. Describe raster animation. [3]

OR

X

1. Describe the steps for animation sequence. [8]
 2. Compare and contrast depth-buffer and depth sort methods. [7]
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